



Auto-ignition control using an additive with adaptable chemical structure. Part I: Development of a kinetic model for 1,3-cyclohexadiene and 1,3,5-hexatriene combustion

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ABSTRACT

1,3-cyclohexadiene (1,3-CHD) can be transformed into *cis*-1,3,5-hexatriene (1,3,5-HT) upon light irradiation, which makes it a potential additive able to change the reactivity of a conventional fuel. This paper presents the development of a detailed chemical kinetic model for the low-temperature (500–1200 K) combustion of 1,3-cyclohexadiene and 1,3,5-hexatriene. Theoretical calculations were performed to compute the thermochemistry of a large number of intermediates involved in the reaction mechanism, and for several kinetic parameters. In particular, the pericyclic reactions of 1,3-cyclohexadiene, linking it to *cis*-1,3,5-hexatriene, were studied theoretically. It was shown that 1,3,5-HT is inherently a secondary molecule of the 1,3-CHD mechanism and a comprehensive set of its oxidation reactions were included. Simulation of literature data (ignition delays and products speciation) measured in rapid compression machines for 1,3-CHD were performed using the newly developed kinetic model. A good agreement with experiments was found, and kinetic analyses highlighted the decomposition mechanism of 1,3-CHD and the most sensitive reactions affecting the auto-ignition delay times. Simulations of *cis*-1,3,5-HT auto-ignition in an RCM were also performed and compared to the ignition behaviour of 1,3-CHD. The simulation results showed that 1,3,5-HT combustion involves an induction period characterized by the predominant formation of 1,3-CHD, whose decomposition starts the radical chain mechanism.

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1. Introduction

A fuel of variable ignition quality can be used to control autoignition timing in a combustion process. Control over autoignition timing of fuel and air mixtures is necessary in combustion processes initiated by autoignition, such as those used in compression ignition engines, but it could also be applied to gas turbine combustors employing auto-ignition as a means of initiating and stabilising combustion [1,2]. In homogeneous charge compression ignition (HCCI) engines, ignition timing depends on the reaction kinetics of fuel and air, and achieving optimum cycle efficiency requires autoignition to occur around the point of maximum compression of the cycle; when the piston is at top-dead-centre (TDC). An HCCI engine needs to have a means of controlling the point of ignition of the fuel and air mixture under different con-

ditions of intake air temperature and boost pressure, fuel and air equivalence ratio, and small variations in fuel quality. The simplest way of controlling ignition timing in an HCCI engine, are physical changes to the engine cycle, such as controlling the intake air temperature through heat transfer [3], or by mechanically adjusting the geometric compression ratio of the engine. It is also possible to control ignition timing chemically, by varying the molecular composition of the fuel and air charge. This can be achieved by using exhaust gas recirculation (EGR) or by blending a fuel of low reactivity with one of high reactivity to the required proportion [4]. Physical changes to the engine cycle, such as increasing the intake air temperature or varying the compression ratio, are often associated with reduced performance of the engine in terms of efficiency or engine load. Chemically controlling ignition timing by means of blending two fuels of different reactivities has the potential to achieve high engine efficiencies, but also has the inherent disadvantage of having to refuel a vehicle with two different fuels [5,6]. In Reactivity Controlled Compression Ignition (RCCI), in-cylinder stratification of fuel reactivity can be used to obtain up to 33% higher efficiencies than comparable diesel engines

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[7–10]. The inconvenience of having to supply two fuels to the engine [5,6] has been solved in previous studies by altering the molecular composition of a fuel prior to combustion taking place, as an additional part of the engine process. This strategy has been demonstrated by catalytic fuel reforming [11], or the simulated catalytic dehydration of an alcohol to an ether [12]. Attempts have also been made to use the concentration of highly reactive gases such as ozone and nitrogen oxides as a means of controlling ignition [13]. Attempts have also been made to use ozone to alter the fuel in liquid phase prior to injection into the engine [14]. The molecular structure of a fuel can also be adapted via irradiation with light [15], to provide control over the autoignition timing. In practice, the photochemically reacted fuel may be stored in a tank. This is necessary because the photochemical conversion process of the fuel is likely to be slower than the transients in engine operation. The fuel composition may be controlled from cycle-to-cycle using a combination of storage tank, engine map, and by continuously measuring the cylinder pressure of the cycles.

In this work, the auto-ignition characteristics of 1,3-cyclohexadiene (1,3-CHD) are studied, using detailed chemical kinetic modelling. This approach is novel in that earlier fuels required the presence of air to form peroxides via photochemical reactions. In the current work, we perform simulations to show that this molecule can be used as an additive whose chemical structure can be modified upon light irradiation, leading to *cis*-1,3,5-hexatriene (1,3,5-HT). The difference of reactivity between 1,3-CHD and 1,3,5-HT is estimated using kinetic simulations.

Photochemical isomerisation of 1,3-cyclohexadiene to *cis*-1,3,5-hexatriene can be used to adapt the molecular structure of the fuel, and thereby influence its ignition reactions with air. This photochemical conversion has previously been documented in detail in the work of Minnaard and Havinga [16].

1,3-CHD is a fuel which can potentially be synthesised from plant oils [17] and it has the practical advantage over previous work [15] that it does not require the presence of air-bound oxygen during its photochemical conversion. To the best of our knowledge, no dedicated kinetic model for the combustion of 1,3-CHD or 1,3,5-HT is available in the literature. In this study, a detailed chemical kinetic model for the combustion of 1,3-cyclohexadiene and 1,3,5-HT is developed and validated against experimental data of the literature. The thermochemistry of a large number of species involved in the mechanism and the kinetics of key elementary reactions linking 1,3-CHD to *cis*-1,3,5-HT are calculated using *ab initio* methods.

2. Kinetic modelling

A detailed chemical kinetic model was developed to simulate the low-temperature combustion (500–1200 K) of 1,3-CHD and 1,3,5-HT. It will be demonstrated that 1,3,5-HT is inherently part of the 1,3-CHD kinetic model. Therefore, 1,3,5-HT mechanism is part of the secondary mechanism of the 1,3-CHD model. The validation of the kinetic model is based on simulations of auto-ignition delays measured in rapid compression machines (RCM). In the literature, such data have been reported by Tanaka et al. [18] for pure 1,3-CHD, at conditions relevant to HCCI combustion: equivalence ratios ranging from 0.2 to 0.5 at pressures around 42 bar and temperatures between 800 and 900 K. Lemaire et al. [19] also reported RCM auto-ignition delays of pure 1,3-CHD in air, for stoichiometric mixtures, between 600 and 900 K and 7–14 bar. These authors also reported concentration-time profiles of the C₆ oxidation products measured during the experiments. In addition, simulations of the experiments of Gulati and Walker [20], conducted in a batch reactor, were also performed. These authors measured the initial products formed in the oxidation of 1,3-CHD added to a slowly reacting H₂–O₂ mixture at 753 K. It should be noted that no experimental data on 1,3,5-HT combustion was found in the literature.

The detailed chemical kinetic model for 1,3-CHD combustion was developed manually by following the rules of the generation algorithm of the EXGAS [21,22] software, an automatic generator of kinetic models. Electronic structure calculations were used to determine unknown decomposition routes and thermokinetic parameters, following a theoretical approach previously developed for a biofuel [23,24]. Following the classic hierarchical structure of combustion mechanisms, the 1,3-CHD kinetic model was built on a core reaction mechanism.

2.1. Core reaction mechanism (C₀–C₆ reaction basis)

The core reaction mechanism includes two reaction bases. The first reaction basis includes all the unimolecular or bimolecular reactions of radicals or molecules featuring less than three carbon atoms. This comprehensive reaction base, denoted as C₀–C₂, is further extended with comprehensive mechanisms for the combustion reactions of propyne, allene, 1,3-butadiene (C₃–C₄ reaction base) [25] and cyclopentadiene and benzene (C₅–C₆). The latter C₅–C₆ mechanism was taken from the updated version included in the 2,5-dimethylfuran combustion kinetic model [26].

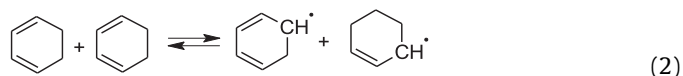
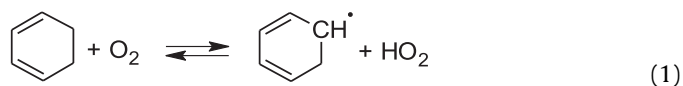
During the course of this study, it was found that the oxidation mechanism of cyclopentadiene in the C₅–C₆ reaction basis had to be extended as cyclopentadiene was found to play a non-negligible role in 1,3-CHD oxidation. Table 1 presents the reactions added to the cyclopentadiene oxidation sub-mechanism. All of them were taken from the JetSurF kinetic model [27] which is based on the USC-Mech II core mechanism [28].

2.2. 1,3-cyclohexadiene sub-mechanism

The 1,3-CHD sub-mechanism is based on the core mechanism described above with primary and secondary mechanisms specific to 1,3-CHD that include unimolecular and bimolecular initiations, pericyclic reactions, H-abstractions, β -scissions, oxidations, additions (including those on O₂ for low-temperature chemistry), isomerisation, cyclic ethers formations, combinations and disproportionations.

2.2.1. Initiations and pericyclic reactions

The initial C–H bond fission in 1,3-CHD, yielding cyclohexadienyl radical, is the only unimolecular initiation considered in the sub-mechanism. It was written as the reverse reaction of cyclohexadienyl combination with H-atom, with a rate constant taken from EXGAS's correlations ($k = 1 \times 10^{14}$ in cm³, mol, cal, s^{–1}). Bimolecular initiations considered in the model involve the reactions of 1,3-CHD with O₂ and with itself:



It can be noted that other reactions are possible, involving vinylic H-atoms or the formation of a non-resonant stabilized cyclohexenyl radical in the self-initiation of 1,3-CHD. In this work, these reactions were neglected because of the high stability of resonant stabilized radicals and the high bond energies of vinylic H-atoms. The rate constants of reactions (1) and (2) were taken from the EXGAS reaction rate rules [22] that estimate activation energies based on enthalpies of reaction. As discussed later in this manuscript, enthalpies of formation of most of the species of 1,3-CHD mechanism were computed using theoretical calculations.

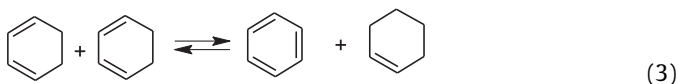
Table 1

Additional reactions for cyclopentadiene oxidation included in the C₀–C₆ core mechanism. Reactions and rate constants are taken from JetSurF [27].

Reaction ^a	k _∞ (A × T ⁿ × exp(–E/RT) in cm ³ , mol, s, cal		
	A	n	E
C5H5x + OOH = C5H5Ox + OH	6.27 10 ²⁹	–4.690	11,650
C5H5x + O ₂ = C5H5Ox + O	7.78 10 ¹⁵	–0.730	48,740
C5H5x + O = C5H5Ox	1.12 10 ^{–12}	5.870	–17,310
C5H5x + O = C5H4Ox + H	5.81 10 ¹³	–0.020	20
C5H5x + O = nC4H5 + CO	3.20 10 ¹³	–0.170	440
C5H5x + OH = C5H4OHx + H	3.51 10 ⁵⁷	–12.180	48,350
C5H5x + OH = C5H5Ox + H	1.36 10 ⁵¹	–10.460	57,100
C5H5x + OH = C5H5OHx	6.49 10 ¹⁷	–0.850	–2730
C5H5x + OH = C5H5OHx	1.15 10 ⁴³	–8.760	18,730
C5H5x + OH = C5H5OHx	1.06 10 ⁵⁹	–13.080	33,450
C5H5x + O ₂ = C5H4Ox + OH	1.80 10 ¹²	0.080	18,000
C5H5OHx + H = C5H4OHx + H ₂	1.20 10 ⁰⁵	2.500	1492
C5H5OHx + OH = C5H4OHx + H ₂ O	3.08 10 ⁰⁶	2.000	0
C5H5Ox + O ₂ = C5H4O + HO ₂	1.00 10 ¹¹	0.000	0
C5H4OHx = C5H4Ox + H	2.10 10 ¹³	0.000	48,000
C5H4Ox + H = CO + C4H5x	4.30 10 ⁰⁹	1.450	3900
C5H4Ox + O = CO + CHO + C3H3	6.20 10 ⁰⁸	1.450	–858
C4H5x + H = C4H6	1.00 10 ¹³	0.000	0
C4H5x + H = C2H4 + C2H2	1.00 10 ¹³	0.000	0
C4H5x + O = CH ₂ CHO + C2H2	1.00 10 ¹⁴	0.000	0
C4H5x + O ₂ = CH ₂ CHO + CH ₂ CO	4.80 10 ¹¹	0.000	19,000
C4H5x = C4H4 + H	3.00 10 ¹²	0.000	52,000
C4H5x = C2H3 + C2H2	2.00 10 ¹²	0.000	58,000

^a Nomenclature of species: cyclopentadienyl (C5H5x), cyclopentadienyloxy (C5H5Ox), 2,4-cyclopentadiene,1-one (C5H4Ox), buta-1,3-dien-1-yl (nC4H5), cyclopentadienyl (C5H4OHx), 1,3-cyclopentadienol (C5H5OHx), cyclobut-2-en-1-yl (C4H5x), 1,3-butadiene (C4H6Z2).

During the course of this study, we explored the feasibility of an alternative pathway in the self initiation of 1,3-CHD. We used theoretical calculations at the CBS-QB3 level of theory [29] and transition state theory, to determine the high-pressure limit rate constant of reaction (3).



In reaction (3), one molecule of 1,3-CHD simultaneously abstracts two hydrogen atoms from another 1,3-CHD molecule, leading to the formation of benzene and cyclohexene. Using the Gaussian09 [30] and ChemRate [31] software, and within the harmonic oscillator approximation, the computed rate constant is $k_3 = 0.194 \times T^{3.573} \times \exp(-21,837/RT)$ in cm³, mol, s, cal units. This reaction features a low activation energy, but also a low pre-exponential factor, and may only be non-negligible under conditions involving high concentrations of 1,3-CHD. The uncertainty of the computed rate constant ranges between factors of 4 at 500 K to 1.6 at 1500 K.

Pericyclic reactions of 1,3-CHD were also included in the reaction mechanism. The only direct pericyclic reaction of 1,3-CHD leads to the formation of *cis*-1,3,5-hexatriene. According to the orbital symmetry rules, the elimination of H₂ in 1,3-CHD is not allowed [32]. However, we calculated the rearrangement of 1,3-CHD into 1,4-cyclohexadiene, through the formation of bicyclo(3.1.0)hex-2-ene, which can either decomposes into benzene and H₂ or yields acetylene and 1,3-butadiene. The potential energy surface (PES) calculated at the CBS-QB3 level of theory is presented in Fig. 1. The computed critical energy for the ring opening of 1,3-CHD into *cis*-1,3,5-HT is 43.6 kcal mol^{–1}. Using quantum chemistry, Ess and Houk [33] reported an estimated experimental barrier for the reverse ring closing reaction of 30.2 kcal mol^{–1} at 0 K. Our calculated value of 28.8 kcal mol^{–1}, is in good agreement with their results. This relatively low value of critical energy for the 1,3-CHD ⇌ 1,3,5-HT inter conversion illustrates the need to include a

detailed model of 1,3,5-HT as a secondary mechanism of 1,3-CHD model.

Using the computed PES and transition state theory, high-pressure limit rate constants were calculated to be included in the 1,3-CHD sub-mechanism. Harmonic oscillator approximation was used and quantum tunnelling was also taken into account using the Eckart method [34]. Several computed rate constants can be compared to experimental measurements of the literature. Ellis and Frey [35] reported rate constants of 6.03 10^{–5} s^{–1} and 9.30 10^{–5} s^{–1} for the unimolecular decomposition of bicyclo(3.1.0)hex-2-ene into 1,4- and 1,3-cyclohexadiene at 600 K, respectively. Our computed rate constants are in good agreement (within less than a factor of 1.7 at 600 K) compared to these data, with calculated values of 1.37 10^{–4} s^{–1} and 1.00 10^{–4} s^{–1}, respectively. The calculated kinetic parameters can be found in the model given in the supplementary material.

2.2.2. H-abstractions

H-abstractions from 1,3-CHD were considered only with small radicals that were found to be in large concentration in the targeted low temperature conditions, and that cannot decompose by unimolecular reactions. For this reason, only H-abstractions by H, OH and HO₂ were considered in the primary mechanism of 1,3-CHD. Because of high bond energies, the abstractions of vinyl H-atoms were neglected. H-abstractions by H atoms were calculated at the CBS-QB3 level of theory with the geometry optimization and frequency calculations steps at the B3LYP/6-311G(d,p) [36,37] replaced by MP2/6-311+G(d,p) [38] level of theory. The substitution of the DFT approach by the post-Hartree-Fock method was found necessary to localize two transition state (TS) geometries. The first one is related to equatorial hydrogen atoms and the second one to axial H-atoms. B3LYP/6-311G(d,p) computations only allowed to successfully localize one TS structure (equatorial H-atom abstraction). The corresponding rate constants (given in Table 2) were computed using transition state theory, using the harmonic oscillator approximation and Eckart transmission coefficients [34]. The kinetic parameters for the H-abstractions by OH and HO₂ were taken from the EXGAS rate rules [22].

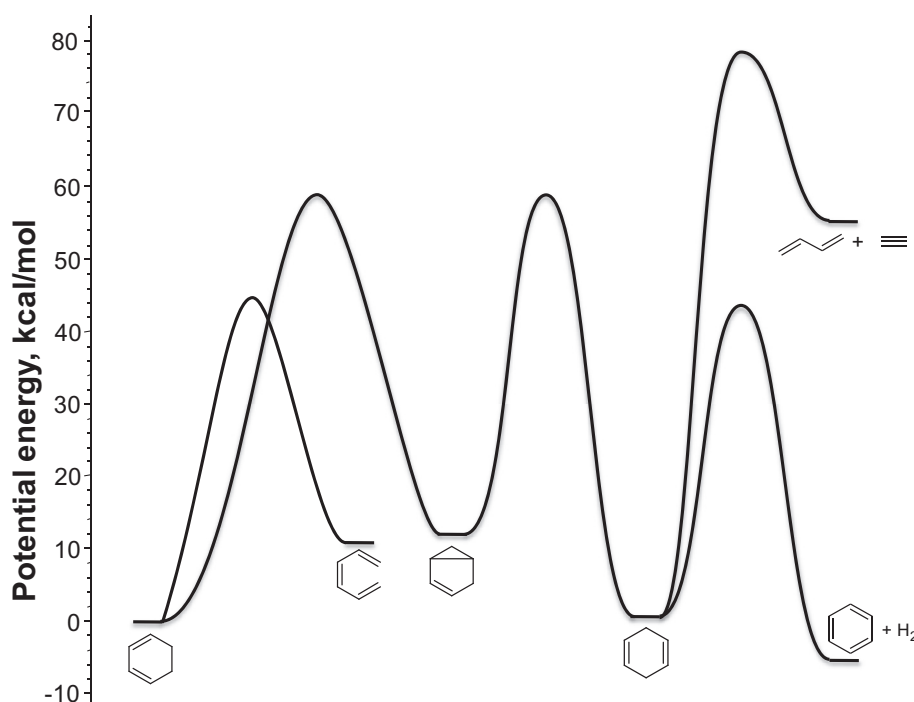


Fig. 1. Potential energy surface of the pericyclic reactions of 1,3-CHD. Energies at 0 K computed at the CBS-QB3 level of theory.

Table 2

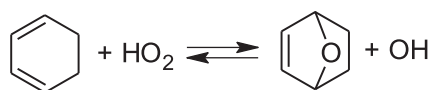
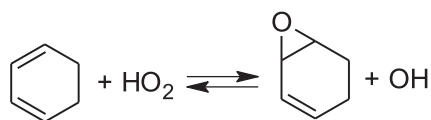
Calculated rate constants for equatorial and axial H-abstractions by H-atom from 1,3-CHD. ($k_{i,\infty} = A_i \times T^{n_i} \times \exp(-E_i/RT)$). Units are in cm^3 , mol, s, cal.

Reaction	$k_{i,\infty}$		
	A_i	n_i	E_i
$\text{xC6H8-1} + \text{R1H} = \text{xC6H7-1} + \text{H}_2$	6.05×10^{06}	2.161	5032
$\text{xC6H8-1} + \text{R1H} = \text{xC6H7-1} + \text{H}_2$	9.67×10^{05}	2.220	1340

2.2.3. Additions

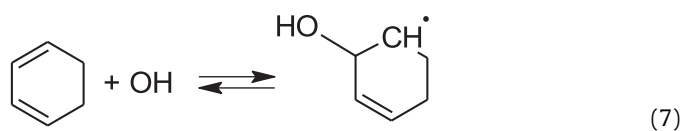
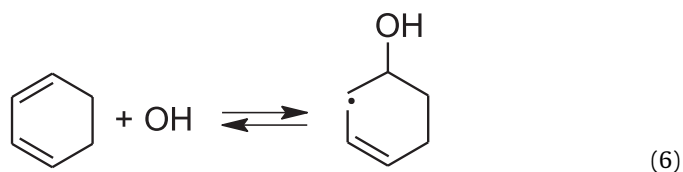
In the 1,3-CHD primary mechanism, several types of additions were taken into account: additions of small radicals on the 1,3-CHD molecules and additions of cyclohexadienyl, hydroxycyclohexenyl and hydroperoxycyclohexenyl radicals on O_2 . The possible decomposition reactions of the adducts include isomerisations, β -scission, disproportionation and cyclic ether formations. All these pathways were written in a systematic and consistent way. This paragraph gives examples for each type of addition considered in the 1,3-CHD primary sub-mechanism.

HO_2 additions on 1,3-CHD were written as a single step reaction yielding cyclic ethers and OH radicals.

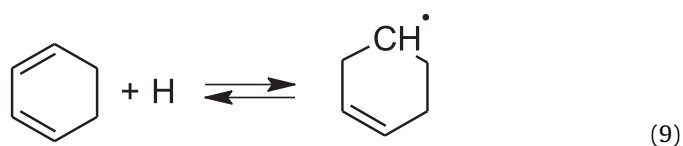
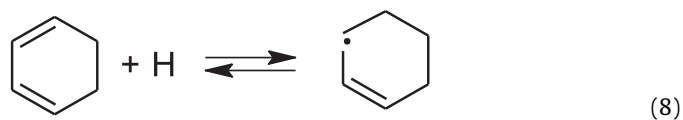


Only two reactions can be envisaged because of the resonance stabilized nature of the hydroperoxycyclohexenyl adduct. The kinetic parameters associated with these reactions ($k_{4,5} = 5 \times 10^{11} \exp(-13,000/RT)$ in cm^3 , mol, s, cal units) were taken from EXGAS rate rules [22], based on the work of Stark [39].

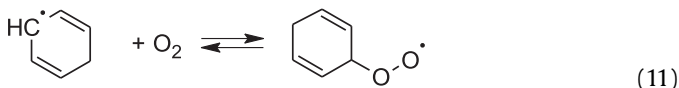
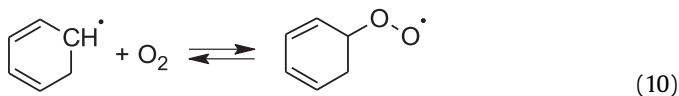
The addition of OH radical on 1,3-CHD was also included in the sub-mechanism. Two types of radicals are created by this addition: a resonance stabilized radical (reaction (6)) and a secondary radical (reaction (7)).



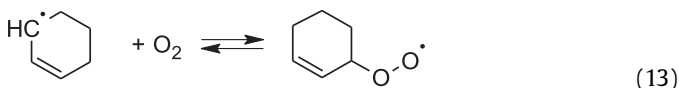
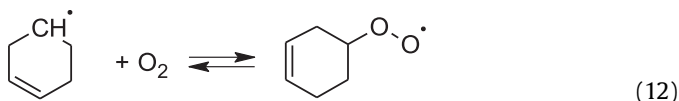
H-atom additions on 1,3-CHD lead to the formation of an alkenyl radical (reaction (9)) or a resonance stabilized cyclohexenyl radical (reaction (8)).



The additions of cyclohexadienyl radicals onto O_2 lead to the formation of two peroxy-cyclohexadienyl ($\text{ROO}\cdot$) radicals, because of the existence of its two mesomeric structures.



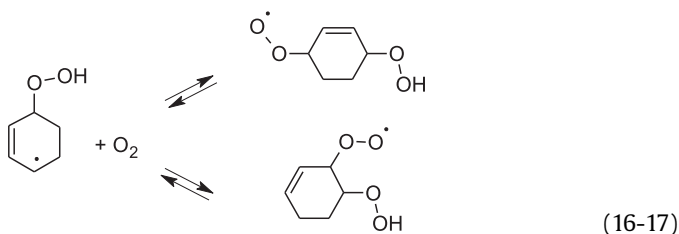
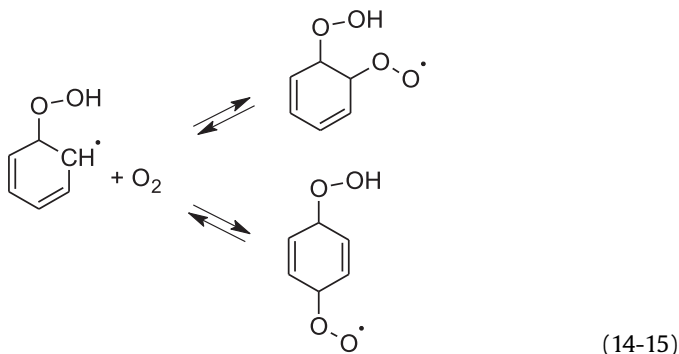
This type of addition on O_2 was also included for cyclohexenyl radicals created in reaction (8) and (9).



Reactions (12) and (13) leads to the formation of singly unsaturated peroxy-cyclohexenyl radicals.

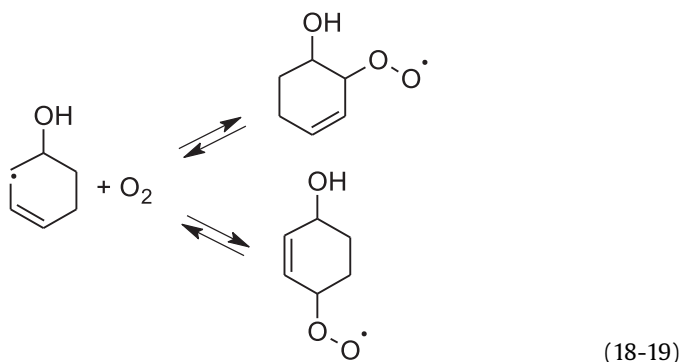
Following the classical low-temperature chemistry schemes, second addition on O_2 was written in the primary mechanism. They involve hydroxycyclohexenyl (reactions (6–7)), hydroperoxy-cyclohexenyl and hydroperoxycyclohexadienyl radicals. The latter radicals are produced by the multiple isomerisations of peroxy radicals from reactions (10) to (13). These reactions will be detailed later in the manuscript.

Additions of resonance stabilized radicals further increase the complexity of the sub-mechanism as their reactions with O_2 leads to two hydroxy- / dihydroperoxy radicals. For example, the addition of the hydroperoxycyclohexadienyl on O_2 leads to two radicals which feature different structural properties regarding the positions of the lateral peroxy/hydroperoxy groups on the cycle and the positions of double bonds.



Reactions (14) and (15) are characteristics of a degenerate mechanism for peroxyradicals formation in resonance stabilized radical compounds. This behaviour also appears for the

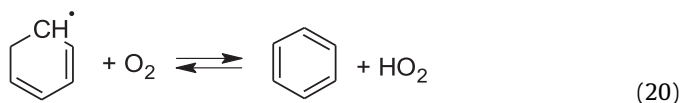
reactions of hydroperoxycyclohexenyl (reactions (16) and (17)) and hydroxycyclohexenyl radicals with O_2 (reactions (18) and (19)).



For all addition reactions on O_2 of 1,3-CHD low-temperature sub-mechanism, the kinetic parameters were taken from EXGAS reactions rate rules for alkyl radicals [22]. The rate constant of the addition of an allylic radical to an oxygen molecule is defined as $k = 1.2 \times 10^{10} \exp(-2300/RT)$ in cm^3 , mol, s, cal, units.

2.2.4. Oxidations

Following the philosophy of our EXGAS automatic generator of combustion kinetic mechanism, we included oxidations reactions in the 1,3-CHD sub-mechanism, which are defined as $\text{R}^\bullet + \text{O}_2 \rightleftharpoons \text{R}' + \text{HO}_2$. $\text{R}^\bullet$ is a radical formed in the primary mechanism and R' is its conjugated unsaturated species. In the sub-mechanism, oxidation reactions were systematically included as elementary processes, which are in competition with addition reactions on O_2 . For example, the oxidation reaction of cyclohexadienyl radical is depicted in reaction (20).

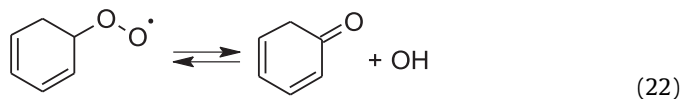
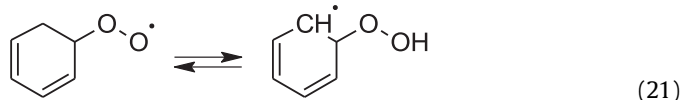


This reaction competes with the addition on O_2 , and yields benzene and HO_2 which are not very reactive species at low temperatures. The associated kinetic data of this specific oxidation was adapted from the core mechanism as it is part of the benzene combustion sub-mechanism. The original rate constant was multiplied by a factor 3.8 to increase the agreement between simulations and experiments. Given the lack of knowledge on this rate constant, this adjustment seems reasonable. Oxidation reactions of cyclohexenyl, hydroxycyclohexenyl, hydroperoxycyclohexadienyl radicals were included in the sub-mechanism, with rate constants taken from EXGAS rate rules [22].

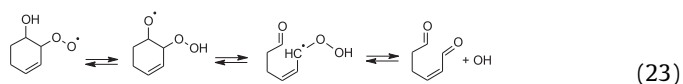
2.2.5. Isomerisations

Internal H-atom transfers were included in the 1,3-CHD sub-mechanism for radicals with sufficient size to allow cyclic transition states with 4, 5, 6 and 7 atoms in the ring. These reactions were considered in the sub-mechanism for alkyl and allyl H-atoms. Isomerisations involving vinyl H-atoms were neglected. Internal H-atom transfers were written for peroxy-cyclohexadienyl, peroxy-cyclohexenyl, hydroperoxy-cyclohexenylperoxy and hydroxyperoxy-cyclohexenyl radicals, produced in reactions (10–11), (12–13), (14–17) and (18–19), respectively.

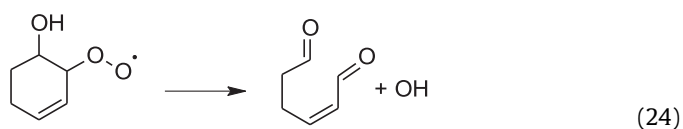
For the peroxy-cyclohexadienyl radical, only two isomerisations can be envisaged following our reaction rules.



In reaction (21), a 5-centered transition state is involved with the transfer of an allyl H-atom. Reaction (22) involves the internal abstraction of an allyl H-atom through a 4-centered TS. Note that it was demonstrated by theoretical calculations that reactions like (22) lead to the formation of a ketone and OH, because the hydroperoxy radicals that should be formed are not stable [40]. In the case of hydroxycyclohexenyl radicals, a different type of H-atom transfer can be written in the sub-mechanism. This reaction involves the transfer of the hydroxyl H-atom and is often referred as the Waddington mechanism [41]. It has been shown that the alkoxy radicals formed in the process are not stable and readily decompose by successive β -scissions, ultimately yielding ketones or aldehydes and an OH radical [42]. In the case of 1,3-CHD, the assumed detailed Waddington mechanism is given in reaction (23).

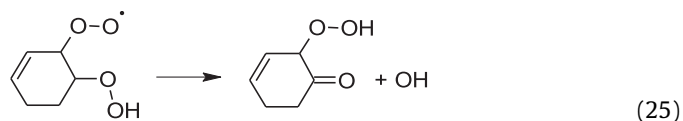


Based on this mechanistic analysis, the Waddington reaction class was assumed to be a one step process leading directly to the bimolecular products (24).



For this specific process, the rate determining step is H-transfer rate constant [42], and it was used systematically for the Waddington reaction class of the sub-mechanism.

Among the isomerisations, another rule was set for hydroperoxycyclohexenylperoxy radicals. To limit the size of the mechanism, the only reactions that were included for these radicals is the one where the peroxy radical centre abstracts the H-atom linked to the carbon atom bearing the hydroperoxide function. Reaction (25) gives an example of the application of this rule to a given radical of this type.



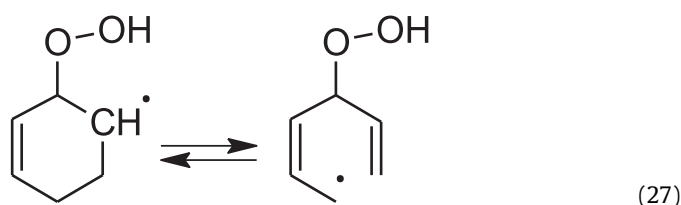
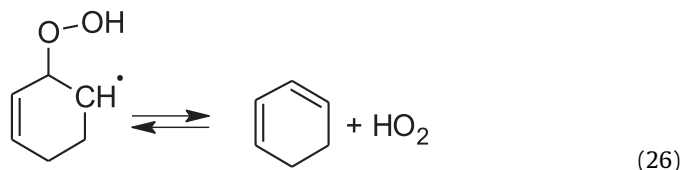
This approach assumes that the abstraction of the H-atom located at the bottom of the hydroperoxide group is the most favoured. This assumption is commonly used in lumping methods to prevent an exponential growth of the mechanism size.

For all isomerisation rate constants, the kinetic correlations developed in EXGAS were used. They are based on three parameters: the nature of the hydrogen atom internally abstracted (alkyl or allyl here), the size of the ring created in the transition state structure, and the variation in the number of internal rotations

between the cyclic TS and the reactant. The first two parameters allow estimating the activation energy and the last one leads to an estimation of the pre-exponential factor.

2.2.6. β -scissions

Given the low-temperature conditions targeted in this work, only C–C, C–O and O–O β -bond fissions were considered in the 1,3-CHD sub-mechanism. In this reaction class, the β -scissions creating a vinyl carbon radical were neglected if other C–C or C–O bond fissions were feasible. These reactions were written for all radicals created in the primary mechanism. Examples of possible β -scissions for one of a hydroperoxycyclohexenyl radical are given below.

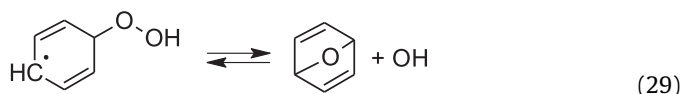
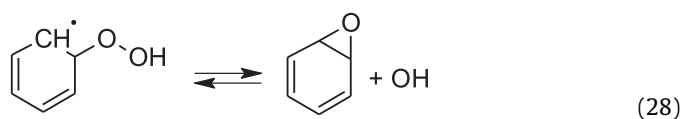


In reaction (26), the β -bond fission occurs at the C–O bond, while in reaction (27) the C–C bond is being broken. Note that the other possible β -scission, not depicted here, would lead to the formation of a vinyl radical. Per our rules of mechanism development, this reaction is neglected in the β -scission reaction class. The rate constants of all β -scissions were taken from EXGAS reaction rate rules [22].

Due to the high-activation energies involved, and the temperatures targeted in this study, C–H β -scissions of free radicals were not taken into account, except for the formation of benzene from cyclohexadienyl radical. Indeed, it will be demonstrated that the C–H β -bond fission in cyclohexadienyl radical, yielding benzene and H-atom, is a key process in the 1,3-CHD combustion mechanism. Therefore the experimental and computational value recommended by Gao et al. [43] was adopted in our kinetic model ($k = 1.20 \times 10^{11} \exp(-21,510/RT)$ in cm^3 , mol, s, cal, units).

2.2.7. Cyclic ethers formation

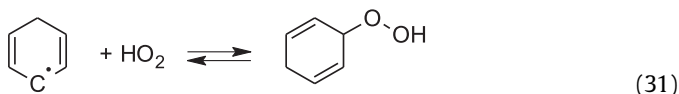
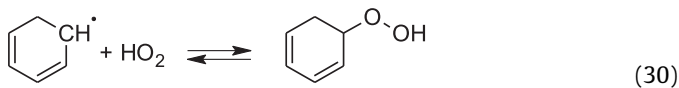
Reactions of hydroperoxycycloalkyl radicals yielding cyclic ethers and OH radicals were included in the sub-mechanism. This reaction class involves the formation of 3-, 4-, 5-membered ether rings onto the cyclohexenyl/cyclohexadienyl radicals. The existence of mesomeric structures in resonance stabilized radicals involves several cyclic ethers pathways as one hydroperoxycycloalkyl radical can lead to the formation of two cyclic ethers.



Reaction rate rules taken from EXGAS were used to determine the associated rate constants.

2.2.8. Terminations

Combinations and disproportionations are the two types of reactions that were included in the termination reaction class. For the resonance stabilized radicals, combinations with HO₂ radicals were systematically included. For example, reactions (30) and (31) were considered in the sub-mechanism as cyclohexadienyl radical was found in high quantities, for a radical species.



Again, the mesomeric forms of this radical increase the complexity of the decomposition routes in this reaction class. Combinations with HO₂ were also included for the following resonance stabilized radicals: cyclohexenyl (in reaction (8)), hydroxycyclohexenyl (in reaction (6)), hydroperoxycyclohexenyl, and hydroxyhydroperoxycyclohexenyl. The rate constant used for these process is the one defined in EXGAS, i.e., $k = 5 \times 10^{12} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$. When preliminary simulations were performed, it was found that the auto-ignition delays of 1,3-CHD were sensitive to the rate constants of reactions (30) and (31). For this reason, the initial rate constant of (30) was multiplied by a factor of 2 while the one of reaction (31) was divided by a factor of 2. Given the uncertainty involved for the kinetic parameters of these reactions, this assumption remains reasonable.

Disproportionations between peroxy radicals and HO₂ were also included. This process is defined as $\text{ROO}^\bullet + \text{HO}_2 \rightleftharpoons \text{ROOH} + \text{O}_2$. This kind of reaction was systematically written for every ROO[•] radicals found in the 1,3-CHD sub-mechanism with a rate constant defined as $k = 2.0 \times 10^{11} \times \exp(1300/RT)$ in cm³, mol, cal, s, per EXGAS reaction rate rules [22].

2.2.9. Secondary mechanism

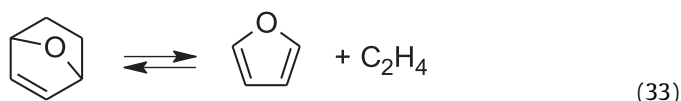
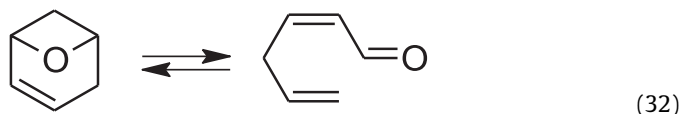
Molecules produced in the primary mechanism, which are not part of the core reaction mechanism (C₀–C₆), are decomposed in the secondary mechanism. In the case of 1,3-CHD, these molecules are cyclic ethers, β-scission products of •QOOH radicals, hydroperoxides, hydroxyhydroperoxides and 1,3,5-hexatriene (1,3,5-HT). To prevent a sharp increase of the 1,3-CHD sub-mechanism size, lumped reactions had been used for some of these compounds, based on the approaches developed in EXGAS.

For all hydroperoxides (hydroperoxycyclohexenyl, hydroperoxycyclohexadienyl, hydroxyhydroperoxycyclohexenyl), the reaction rule adopted assumes an explicit O–O bond breaking, $\text{ROOH} \rightarrow \text{RO}^\bullet + \bullet\text{OH}$, with an associated rate constant of $k = 1.5 \times 10^{16} \times \exp(-43,000/RT)$ in cm³, mol, cal, s units. The produced alkoxy radicals RO[•] are then decomposed by successive β-scissions. The first β-scissions opens the alkoxy radical, yielding an isomer of RO[•], which is written in the mechanism with a rate constant taken from EXGAS's correlations. The decomposition reactions of the linear isomers of RO[•] are then lumped assuming successive β-scissions into small fragments that further react in the core mechanism.

Decomposition reactions of cyclic ethers were also lumped following the reaction rules developed in EXGAS. This approach assumes that after H-abstraction of the most labile hydrogen atom, the cyclic ether radical formed readily decomposes by successive

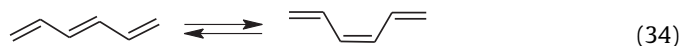
β-scissions. Therefore, H-abstractions from cyclic ethers by H, OH, and HO₂ were assumed to directly yield small fragments that react in the core mechanism. It can be noted that the nature of each hydrogen atom on a given cyclic ether allowed us to determine the most likely H-atom to be abstracted. The following ranking of stability was used, from the lowest to the highest bond energy: $H_{\text{resonance-stabilized}} > H_{\text{tertiary}} > H_{\text{secondary}}$.

Lumped reactions were found to be not suitable for two cyclic ethers produced in the mechanism. For these species, pericyclic reactions can be envisaged and therefore lead to a more complex decomposition scheme.



Reaction (32) yields an unsaturated aldehyde while reaction (33) produces furan and ethylene. For these two pericyclic reactions, the high-pressure limit rate constant was theoretically calculated using CBS-QB3 calculations and the harmonic oscillator approximation. The computed kinetic parameters are $k_{32} = 7.28 \times 10^{12} \times T^{0.484} \times \exp(-44,051/RT)$ and $k_{33} = 3.78 \times 10^{12} \times T^{0.563} \times \exp(-33,606/RT)$ in s⁻¹ and cal/mol units. The decomposition reactions of 2,5-dihexenal were lumped in three reactions assuming that the H-abstraction from the aldehydic H-atom by H, OH and HO₂ directly yields CO and C₅H₇, which are species included in the core mechanism. A detailed mechanism for furan combustion was added to the 1,3-CHD sub-mechanism. It was taken from a previous work performed in our laboratory [44]. H-abstraction reactions of the cyclic ethers of reactions (32) and (33) were explicitly written for the most labile H-atoms, and several successive β-scission reactions were also included for the formed cyclic ether radicals.

The oxidation mechanism of 1,3,5-HT was developed in a comprehensive way, following the same methodology as the one detailed above for 1,3-CHD. Because of the vinyl nature of all C–H and C–C bonds in 1,3,5-HT and its symmetry, its combustion sub-mechanism is more simple than the 1,3-CHD one. In this mechanism, two isomers of 1,3,5-HT were considered. The species named "1C6H8" in the mechanism represents 1,3,5-HT in the *cis* conformation, while "HXT_trans" is the same molecule in the all *trans* conformation. The inter-conversion between these two conformers is included in the mechanism through the following reaction,



where the thermochemical data were calculated using the computational method described above and the rate constant was taken from Orchard and Thrush [45], i.e., $k_{34} = 4.47 \times 10^{12} \times \exp(-43,373/RT)$ in cm³, mol, s⁻¹, cal, units. Distinguishing between the two conformers is critical for their combustion kinetics as the *cis* conformer can cyclize into 1,3-CHD through a pericyclic reaction (see Fig. 1), with a low energy barrier (28.8 kcal/mol at the CBS-QB3 level of calculation), while this reaction is forbidden in the *trans* conformation. Therefore, it can be expected that *cis*-1,3,5-HT and 1,3-CHD will share a large extent of common combustion mechanisms. It can be also

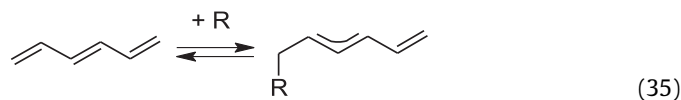
noted that the ignition delay of 1,3-CHD measured in a RCM by Tanaka et al. [18] (4.50 ms) is shorter than those of cyclohexane (5.09 ms) and cyclohexene (7.90 ms) under the same conditions ($\Phi = 0.4$, $T_i = 318$ K, $P_i = 0.1$ MPa, Compression Ratio = 16). According to Tanaka et al., these conditions lead to temperature and pressure at the end of compression of 827 K and 41.6 bar. 1,3-CHD therefore presents a surprisingly high reactivity, and its relatively easy formation from *cis*-1,3,5-HT will probably promote the reactivity in comparison with the case where its production is more difficult (*trans*-1,3,5-HT). Our simulation results, presented below, will concur with these assumptions.

The *cis-trans* inter-conversion of 1,3,5-HT was added to the block of pericyclic reactions of the 1,3-CHD combustion sub-mechanism. For both 1,3,5-HT conformers, bimolecular initiations with O_2 were included in the mechanism. These reactions involve the abstraction of a vinyl H-atom by O_2 with rate constants taken from EXGAS rate rules [22]. Additions of H, OH and HO_2 radicals on 1,3,5-HT were also included in the model. Three OH additions and two HO_2 additions were included, yielding $\bullet ROH$ radicals and cyclic ethers and OH radicals, respectively. In $\bullet ROH$ radicals, isomerisations involving a non vinyl H-atom and β -scissions were considered in the mechanism. Additions of $\bullet ROH$ radicals on O_2 were also considered in the mechanism as well as isomerisations and Waddington mechanisms of the $\bullet OOROH$ radicals. $HOOR\bullet OH$ radicals created in the isomerisations were decomposed into hydroxyl cyclic ether and OH. β -scissions of these radicals were neglected because of the vinyl nature of the produced radicals.

Following the mechanism development for 1,3-CHD, H-abstractions from 1,3,5-HT by H, OH and HO_2 were included in the mechanism. Decomposition by β -scissions and oxidations of the radicals $R\bullet$ created in H-abstractions were neglected because they involve processes with high activation energies compared to competitive pathways. This assumption is acceptable for the simulations of low-temperature conditions. $R\bullet$ additions on O_2 were included in the sub-mechanism, but unimolecular decomposition routes of $ROO\bullet$, initiated by isomerisations, were neglected because they involve the transfer of a vinyl H-atom and internal rotations over unsaturated bonds to form the transition states. The main decomposition route $ROO\bullet$ radicals was considered through their disproportionation reactions with HO_2 , in the termination steps. $R\bullet$ radicals are also decomposed in the termination reactions block where they react with HO_2 by combination, yielding secondary hydroperoxides. Note that the resonance stabilized nature of the created radicals leads to a number of $ROO\bullet$ and hydroperoxydes greater than the initial number of radicals.

The secondary mechanism of 1,3,5-HT was developed in the same way as the 1,3-CHD one. Lumped reactions were used for the decomposition of hydroperoxides and cyclic ethers, using the same rules described in details in the 1,3-CHD sub-mechanism section of this article.

Bimolecular initiations, H-abstractions, OH, H and HO_2 additions were written for both *trans* and *cis* configurations of 1,3,5-HT. For simplicity, these reactions were defined as yielding the same set of products in the mechanism. This assumption can be made when the decomposition pathways of the products formed do not involve internal rotations over the central double bond of 1,3,5-HT. For example, *cis* and *trans* $R\bullet$ radicals follow the same decomposition routes: $R\bullet + O_2 \rightleftharpoons ROO\bullet$, $ROO\bullet + HO_2 \rightarrow ROOH + O_2$, and $ROOH \rightarrow RO\bullet + OH$. Within this approach the final decomposition products are either identical or *cis-trans* conformers that were lumped in a single species. It can also be noted that some additions of a $R\bullet$ radical on a double bond in a conjugate alkene can create a 3 electrons bond instead of the central π bond which makes the rotation around it less difficult, as illustrated in scheme (35).



Bounaceur et al. [46] calculated a rotational barrier of about 15 kcal mol⁻¹ for this 3 electrons rotational process in resonance stabilized pent1-en-3yl radicals, at the CBS-QB3 level of theory.

The addition of a $R\bullet$ radical on the central double bond of 1,3,5-HT even transform the double bond into a simple bond for which the internal rotation is easy. This phenomenon can scramble the initial distinctions between *cis* and *trans* conformations.

All thermochemical data of the 1,3,5-HT sub-mechanism were calculated with the THERGAS software [47]. All rate constants were taken from EXGAS reaction rate rules [22].

2.2.10. Thermochemistry

Preliminary simulations showed that auto-ignition delays of 1,3-CHD were sensitive to the thermochemistry of reactions involving additions of radicals on O_2 . The calculations of thermochemical data of cyclic species with the Benson method were also found to be uncertain, because several groups and intra-molecular interactions were not well defined. For this reason, we calculated the thermochemical data of 1,3-CHD cyclic species using a theoretical approach. Quantum chemistry calculations were performed at the CBS-QB3 level of theory. Vibrational partition functions of compounds featuring internal rotations were corrected using the 1D-HRU method proposed by Vansteenkiste et al. [48] which is implemented in our, in-house, THERMROT code [42]. This method allows treating the coupling between torsional motions using the partition function expressed as:

$$Q_{1D-HRU} = Q_{HO} \prod_{j=1}^N \frac{q_{1D-HR,j}}{q_{HO,j}}$$

where Q_{HO} is the full partition function in the harmonic oscillator approximation. $q_{1D-HR,j}$ is the partition function of the torsional mode j , calculated using the 1D-HR method. $q_{HO,j}$ is the harmonic partition function for the same torsional mode but computed from the second derivative of the calculated torsional potential $V(\theta)$ with respect to the dihedral angle θ . In this work relaxed scans were performed at the B3LYP/cbsb7 level of calculations to determine $V(\theta)$ for all internal rotations.

Using corrected partition functions $S^\circ_{298\text{ K}}$ and $C_p^\circ(T)$ were calculated using standard statistical thermodynamic theory. Atomization reactions and corrected partition functions were used to calculate standard enthalpies of formation at 298 K. These data were included in the 1,3-CHD sub-mechanism as NASA polynomials. Table 3 presents the results of the thermochemistry calculations for a selection of cyclic species.

Table 3 features computed thermochemical data of about 40 species out of more than a hundred that were calculated in this work. The full list of species for which thermodynamic data were computed can be found in supplemental material. The experimental value of enthalpy of formation of cyclohexadienyl radical was taken from the experimental work of Gao et al. [43].

3. Validation of the kinetic model

3.1. Simulations of 1,3-CHD auto-ignition data in rapid compression machines

Validations of the 1,3-CHD detailed kinetic model were performed against ignition delay data measured by Tanaka et al. [18], for a single condition, and Lemaire et al. [19] for temperatures

Table 3
Computed thermochemical properties of selected cyclic species of the 1,3-CHD sub-mechanism (see text for computational details).

	$\Delta_f H_{298}^\circ$ (kcal mol ⁻¹)	S_{298}° (cal mol ⁻¹ K ⁻¹)	$C_p(T)$ (cal mol ⁻¹ K ⁻¹)						
			300 K	400 K	500 K	600 K	800 K	1000 K	1500 K
	10.5	89.56	32.18	42.18	50.47	57.26	67.16	73.46	82.99
	39.4	87.06	30.07	37.85	44.50	50.14	58.77	64.81	72.85
	40.5	88.27	29.48	37.50	44.30	50.2	58.72	64.81	72.89
	27.7	93.82	33.91	42.36	49.72	56.05	65.91	72.83	82.10
	26.4	93.69	33.45	42.13	49.63	56.04	65.94	72.86	82.10
	11.0	91.99	32.89	41.70	49.30	55.78	65.76	72.70	81.97
	30.6	90.33	31.07	38.91	45.56	51.12	59.50	65.19	72.70
	-0.1	75.36	24.79	33.65	41.16	47.40	56.58	63.28	72.49
	5.5	73.24	23.88	33.16	40.94	47.33	56.71	63.33	72.53
	-5.8	71.29	23.45	33.01	40.93	47.38	56.75	63.32	72.48
	24.3	71.95	23.11	31.10	37.64	42.92	50.57	55.89	63.23
	7.3	90.21	31.47	39.60	46.66	52.70	62.06	68.54	77.14
	6.1	88.74	31.59	40.01	47.16	53.17	62.32	68.68	77.17
	-24.6	94.01	33.95	44.08	52.57	59.59	69.94	76.63	86.82
	-7.5	102.60	41.68	51.49	59.69	66.49	76.66	83.66	92.96
	-10.2	105.78	39.31	48.88	57.15	64.21	75.07	82.58	92.41
	-9.6	105.59	39.13	48.83	57.16	64.24	75.08	82.57	92.40
	-50.5	89.13	33.59	42.89	50.99	57.93	68.59	75.84	84.32
	-50.3	94.31	34.80	43.42	50.88	57.24	67.07	73.89	82.80
	-43.4	94.47	36.03	44.80	52.16	58.30	67.62	74.33	83.11
	-6.5	82.74	29.90	37.85	44.79	50.78	60.22	67.02	76.42
	-27.4	94.84	35.99	45.51	53.53	60.23	70.36	77.44	87.08
	-50.5	89.13	33.59	42.89	50.99	57.93	68.59	75.84	84.32

(continued on next page)

Table 3 (continued)

	$\Delta_f H_{298}^\circ$ (kcal mol ⁻¹)	S_{298}° (cal mol ⁻¹ K ⁻¹)	$C_p^\circ(T)$ (cal mol ⁻¹ K ⁻¹)						
			300 K	400 K	500 K	600 K	800 K	1000 K	
	44.2	77.80	25.44	33.59	40.39	45.97	54.23	60.08	68.19
	41.9	78.87	25.43	33.52	40.29	45.87	54.11	59.94	68.06
	-12.3	99.40	40.42	49.55	57.05	63.17	72.21	78.53	87.23
	-28.9	96.72	39.72	48.86	56.41	62.62	71.84	78.30	87.11
	-37.9	78.76	29.94	39.53	47.34	53.65	62.81	69.01	77.59
	-46.3	78.51	29.85	39.66	47.52	53.79	62.78	68.90	77.47
	-48.7	110.68	46.76	56.70	64.93	71.69	81.71	88.59	97.78
	-47.8	110.78	46.28	56.55	64.98	71.84	81.89	88.75	97.87
	-44.2	106.64	41.74	51.57	60.12	67.45	78.76	86.51	96.73
	-43.3	107.02	41.21	51.21	59.86	67.25	78.61	86.42	96.69
	-83.2	109.25	48.84	59.31	67.98	75.09	85.58	92.66	102.13
	-83.4	106.40	48.18	59.62	68.72	75.93	86.13	92.94	102.30
	3.4	82.91	32.05	41.06	47.97	53.26	60.57	65.78	73.22
	-4.0	84.80	30.89	38.97	45.76	51.41	59.89	65.74	73.38
	-1.0	76.14	24.78	33.54	41.03	47.29	56.62	63.27	72.52
	-41.4	81.43	30.12	39.48	47.32	53.81	63.37	69.50	77.85
	-35.9	78.66	28.94	38.69	46.91	53.73	63.74	69.98	78.13

ranging between 675 and 860 K at pressures around 8 and 13 bars. Both experimental data sets were measured in RCMs with flat pistons and can suffer from the influence of the roll-up vortex on the homogeneity of the core region. Mittal and Chomier [49] discussed in details the interpretation of experimental data from RCM without creviced pistons. They compared CFD and homogeneous modelling results and demonstrated that the discrepancy in the ignition data are expected to be moderate under conditions of mild NTC behaviour or pressures above 8 bar. The same conclusion was reached by Griffiths et al. [50] who performed CFD modelling of cyclohexane auto-ignition in the Lille's RCM. Indeed, Lemaire et al. [19] measured auto-ignition delays for cyclohexane, cyclohexene and 1,3-cyclohexadiene in the same study. The experimen-

tal data used in this study for the validation of the kinetic model fulfil these conditions of moderate non-homogeneities. However, it should be noted that the homogeneous modelling of such data may induce an additional uncertainty in the comparisons between experiments and simulations.

Tanaka et al. [18] reported auto-ignition delays of pure hydrocarbon fuels and mixtures in a Rapid Compression Machine for Homogeneous Charge Compression Ignition (HCCI) conditions, i.e., fuel lean and ≈ 40 atm pressure [18].

1,3-CHD ignition delay in this RCM was simulated using the Chemkin-Pro software [51]. Note that the absolute tolerance criterion of the solver was set to 1×10^{-17} to avoid convergence issues. The compression stage and the heat loss in the RCM were included

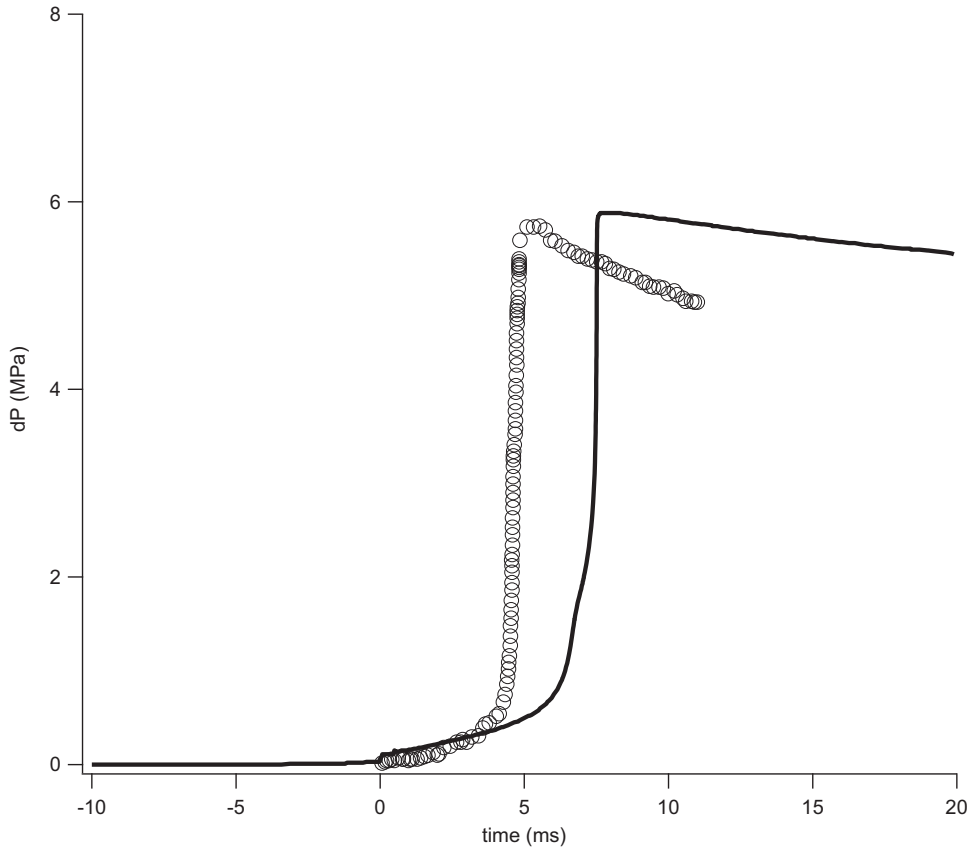


Fig. 2. dP as a function of time in a RCM for a 1,3-CHD/air mixture with an equivalence ratio of 0.4. Initial temperature and pressure of 318 K and 0.1 MPa with a compression ratio of 16. The pressure at the end of compression (P_c) measured by Tanaka et al. [18] is 41.6 bar, temperature at the end of compression calculated from measured P_c and isentropic relationships (T_c) is 827 K. Symbols: experimental data of Tanaka et al. [18]; lines: simulations.

in the simulations, following the model proposed by Tanaka et al. [52], applied to the characteristics of their RCM.

In their heat transfer model, the "adiabatic core" effective volume is calculated as:

$$V_c = V + \delta A = \pi (r + \delta)^2 \times (h + 2\delta) \quad (1a)$$

where V is the actual cylinder volume of the RCM, δ is the displacement thickness of the thermal boundary layer, A is the wall area of the cylinder, r is the bore radius, and h is the actual cylinder height:

$$h = \begin{cases} h_2 + d - u_p t & : t \leq 0 \\ h_2 + d \left(1 - \frac{u_p t}{2d}\right)^2 & : 0 < t \leq \frac{2d}{u_p} \\ h_2 & : \frac{2d}{u_p} < t \end{cases} \quad (2a)$$

with h_2 being the clearance height between piston head and cylinder head, u_p is the constant velocity of the piston, and d the stopping distance such as

$$d = u_p t_s / 2 \quad (3a)$$

where t_s is the time between the inflection point and the peak compression pressure. In Tanaka et al. [52] heat transfer model, the thickness of the boundary layer is defined as:

$$\delta = \sqrt{\alpha_u \cdot t} \quad (0 < t) \quad (4a)$$

with α_u is the thermal diffusivity that is fitted with the compression curves for an inert gas mixture featuring the same specific heat ratio as the air/fuel mixture studied. In our work, d in Eq. (3a) was calculated to be 0.002328 m, based on $u_p = 9.7$ m/s and $t_s = 0.48$ ms. Simulations were performed with an inert

CO₂/air mixture to determine the value of α_u that fitted the inert mixtures experiments of Tanaka et al. [18]. A value of 22.5×10^{-6} m²/s was found for α_u . Following the assumption made by Andrae [53], the insertion of second inflection point, caused by the pressure rise due to combustion, was neglected in the heat transfer model of Tanaka et al. Comparisons between experiments and simulations for the pressure-time profile of 14% of CO₂ in air using the heat transfer model defined above, can be found in the supplementary material. Tanaka et al. [18] measured pressure-time profiles for fuel/air mixtures and CO₂/air mixtures with the same specific heat ratios. They reported pressure differences (dP) between the two mixtures for each condition and defined ignition delays as the time at which dP reached 20% of the dP_{max} point. In our modelling study, we followed this definition and simulated combustion for fuel/air mixtures and for CO₂/air mixtures.

Figure 2 presents a comparison between experiments and simulations of the pressure difference between a 1,3-CHD/air mixture and an inert CO₂/air mixture as a function of time measured by Tanaka et al. [18].

It can be seen from Fig. 2 that the 1,3-CHD kinetic model is able to satisfactorily reproduce the experimental reactivity. From the simulation plot, an ignition delay of 6.5 ms is predicted, following the definition of Tanaka et al. [18] (20% of dP_{max}). This value is in reasonable agreement with the experimental ignition delay of 4.5 ms reported by these authors.

Simulations of RCM data of Lemaire et al. [19] were performed using a closed homogeneous reactor with a volume of 38 cm³. Silke et al. [54] simulated cyclohexane data measured in the same RCM and investigated the effect of different heat loss models

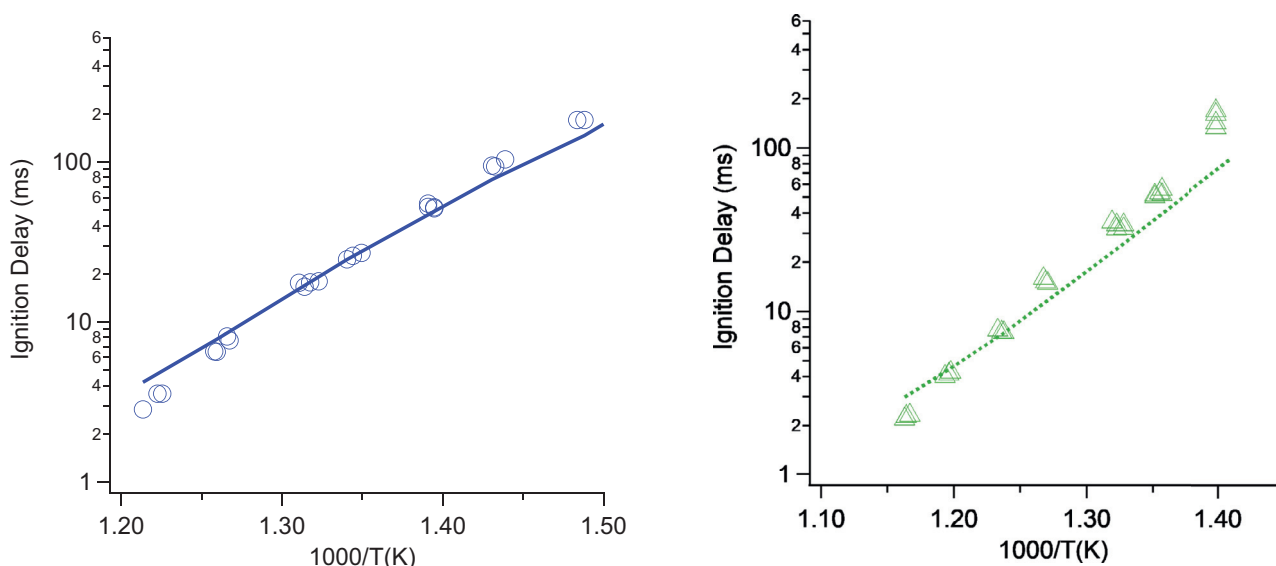


Fig. 3. Ignition delays of 1,3-CHD/O₂/inert mixtures in a RCM ($\Phi = 1$). Symbols are experimental data measured by Lemaire et al. [19] (blue circles: ≈ 12.8 bar; green triangles ≈ 8.3 bar). Lines represent simulated ignition delays (solid: ≈ 12.8 bar; dotted: ≈ 8.3 bar). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

on the simulations. They demonstrated that heat loss inclusion could be important for the lowest temperatures of Lemaire et al. [19] data. They reported simulated ignition delays that were $\approx 30\%$ longer between 650 and 700 K with a heat loss model similar to the one proposed by Tanaka et al. [52], compared to the adiabatic model. Since very few data were measured below 700 K for 1,3-cyclohexadiene, no heat loss model was considered in our simulations. It can be noted that the authors from Lemaire et al. [19] also performed a modelling study of their data (only for cyclohexene), in a subsequent article, where they used an adiabatic closed homogeneous reactor [55]. Actual reactive experimental mixture compositions and pressures were used in the simulations since Lemaire et al. [19] had to vary the inert composition (Ar / N₂) to measure ignition delays at different temperatures. The composition of the mixtures were 2.56% 1,3-CHD / 20.47% O₂ / 76.97% inert (N₂ / Ar). Two sets of data were reported by these authors:

- pressures at top dead centre (P_{TDC}) ranging from 11.3 to 14.3 bar (mean pressure of 12.8 bar),
- and $7.8 \leq P_{TDC} \leq 9.2$ bar (mean pressure of 8.3 bar) data.

These data are presented in Fig. 3, along with simulation results.

In Fig. 3, simulated ignition delays were defined as the temperature inflection point given by Chemkin Pro. In order to check the validity of this criterion, the ignition delay defined by the peak in the OH profile was also computed. Identical simulation results were found using these two criteria. Simulated ignition delays displayed in Fig. 3 show a very good agreement with experimental data above 700 K. Below this temperature, simulated ignition delays are underestimated for the lowest pressures. This can be linked to the adiabatic reactor model used in this modelling work as previously mentioned in this manuscript. The scatter of 8.3 bars data (green triangles in Fig. 3) at ≈ 710 K also shows that for longer ignition delays, the reproducibility of the RCM experiments is more difficult to achieve.

A very interesting point in the work of Lemaire et al. [19] lies in their detailed species measurements that allow a further validation of the 1,3-CHD kinetic mechanism. These authors reported quantifications of intermediate products formed before auto-ignition. In the case of 1,3-cyclohexadiene, they measured the

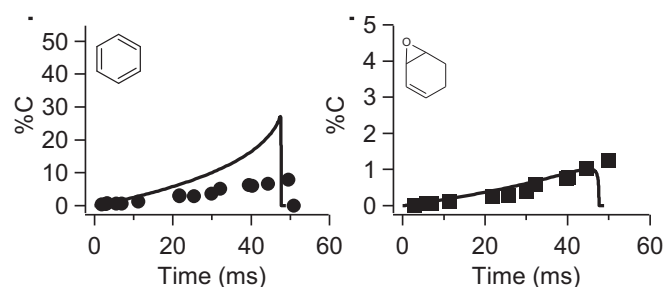


Fig. 4. Concentration-time profile of benzene and 1,2-epoxycyclohex-3-ene in a RCM (pressure and temperature at top dead centre (T_{TDC}): 10 bar and 727 K, respectively). Symbols: experiments [19]; solid lines: simulations.

concentration time profiles of benzene, 1,4-epoxycyclohex-2-ene, 1,2-epoxycyclohex-3-ene and cyclopentenyl-carboxaldehyde.

The last species of this list is not produced in our mechanism. In Lemaire et al. [19] experiments, it accounts for less than 0.2% C (percent carbon in the fuel) of the measured product selectivity. These authors suggested that its formation is due to the decomposition of 1,2-epoxycyclohex-3-ene. This hypothesis could not be tested using our kinetic model as the decomposition mechanism of this cyclic ether is lumped per our mechanism development rules presented above. Figure 4 displays a comparison between simulated and experimental C6 species profiles reported by Lemaire et al. [19].

Benzene is the major product experimentally quantified in the oxidation of 1,3-CHD. It should be noted that Gulati and Walker [20] performed experiments on 1,3-cyclohexadiene slowly reacting in H₂-O₂ mixtures at 753 K, and found that benzene was the only major initial product from 1,3-CHD. Our simulations agree with the conclusions of Gulati and Walker [20] and the data of Lemaire et al. [19], as shown in Fig. 4. It can be noted that the literature simulations of these RCM speciation data always predict a sharp increase of the concentration profile before auto-ignition [54,55] that is not featured in the experimental data. This is probably due to the difficulty of quantifying large concentration variations within a very short amount of time.

Lemaire et al. [19] also reported a species profile for 1,4-epoxycyclohex-2-ene. The formation mechanism of this cyclic

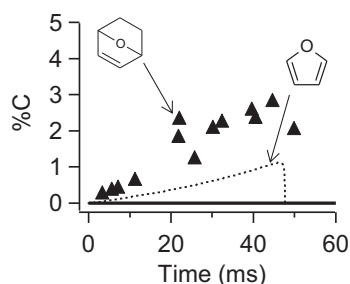


Fig. 5. Concentration-time profile of 1,4-epoxycyclohex-2-ene in a RCM (pressure and temperature at top dead centre of 10 bar and 727 K, respectively). Symbols: experiments [19]; solid line: simulations; dotted line: simulated furan profile.

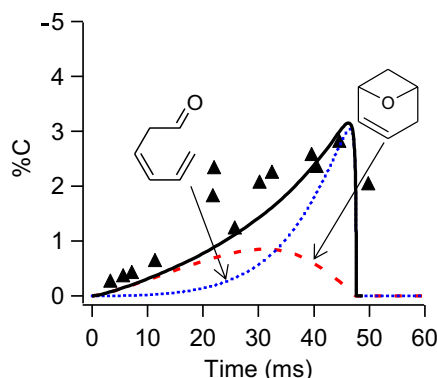
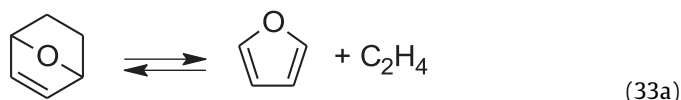


Fig. 6. Concentration-time profile of 1,4-epoxycyclohex-2-ene in a RCM (pressure and temperature at top dead centre of 10 bar and 727 K, respectively). Symbols: experiments [19]; solid line: simulations (dashed red: 6-oxabicyclo [3.1.1] hept-2-ene; dotted blue: hexa-3,5-dienal; black: sum of red and blue profiles). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ether is included in our mechanism. However, according to our computations, it is not predicted to be a stable intermediate. Figure 5 presents the experimental and simulated species profile for this compound.

Using our detailed kinetic model, the 1,4-epoxycyclohex-2-ene profile remains close to 0. This is explained by an easy unimolecular decomposition reaction that was characterized in our study. The retro-Diels-Alder reaction



yielding furan and ethylene was included in the mechanism. The theoretical calculation shows that reaction (33a) features a low activation energy of 33.6 kcal/mol and is therefore expected to occur easily under the temperature conditions of Lemaire et al. [19]. Reaction flux analyses will show that reaction (33a) explains the negligible profile of 1,4-epoxycyclohex-2-ene and confirm that furan solely comes from its decomposition. For comparison, the simulated concentration of furan is included in Fig. 5 to give an indirect information on 1,4-epoxycyclohex-2-ene.

Using simulation results, we examined the mole fraction profiles of other possible cyclic ethers, that are isomers of 1,4-epoxycyclohex-2-ene. Among the possible isomers, it was found that 6-oxabicyclo[3.1.1]hept-2-ene and hexa-3,5-dienal could account for the concentration time profile measured by Lemaire et al. [19], attributed to 1,4-epoxycyclohex-2-ene. Figure 6 presents the simulated concentration profiles of these two isomers, compared to experimental data.

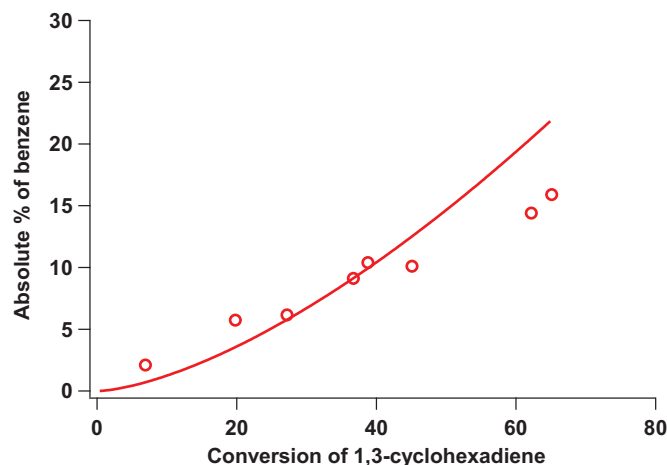
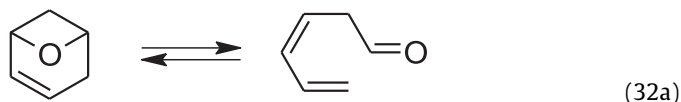


Fig. 7. Benzene formation as a function of 1,3-cyclohexadiene conversion at 753 K. Composition of the reacting mixture: $\text{H}_2 = 140$, $\text{O}_2 = 70$, $\text{N}_2 = 287.5$, 1,3-CHD = 2.5 Torr. Symbols: experimental data of Gulati and Walker [20]; solid line: simulations. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

It can be seen from Fig. 6 that the sum of the two isomers 6-oxabicyclo[3.1.1]hept-2-ene and hexa-3,5-dienal concentration-time profiles allows to simulate the experimental profile reported for a similar isomer. As for 1,4-epoxycyclohex-2-ene (reaction (33a)), a pericyclic reaction was identified for 6-oxabicyclo[3.1.1]hept-2-ene leading to hexa-3,5-dienal.



3.2. Simulations of the oxidation of a 1,3-CHD/ H_2/O_2 mixture

Gulati and Walker [20] experimentally studied the addition of 1,3-cyclohexadiene to a slowly reacting $\text{H}_2\text{-O}_2$ mixture at 753 K. In their study, they measured the initial product yields as a function of 1,3-CHD conversion in a cylindrical Pyrex vessel, coated with boric acid. The kinetic experiments were carried out using a mixture containing 2.5 Torr of 1,3-CHD, with 70 Torr of O_2 , 140 Torr of H_2 and 287.5 Torr of N_2 . Benzene was found to be the only major product from 1,3-CHD [20]. Simulations were performed under the same conditions, in a batch adiabatic reactor. The modelling results are compared to the experiments in Fig. 7.

The simulated benzene profile as a function of 1,3-CHD conversion is able to satisfactorily reproduce the experimental data of Gulati and Walker [20]. This comparison suggests that, for this condition, the kinetics of 1,3-CHD consumption and the yield of benzene, which is the major primary product (see 3.3 section below), are well captured by the primary mechanism of the kinetic model.

3.3. Kinetic analysis of 1,3-CHD combustion

Since our validated 1,3-CHD kinetic model is the first one proposed in the literature, it is interesting to examine the detailed chemistry occurring under combustion conditions. Figure 8 presents a flux analysis performed for Lemaire et al. conditions [19], for which C_6 intermediate profiles were quantified (Figs. 4–6).

It can be observed from Fig. 8 that the decomposition routes of 1,3-CHD under low-temperature combustion conditions are complex. Three types of elementary reactions are highlighted in the

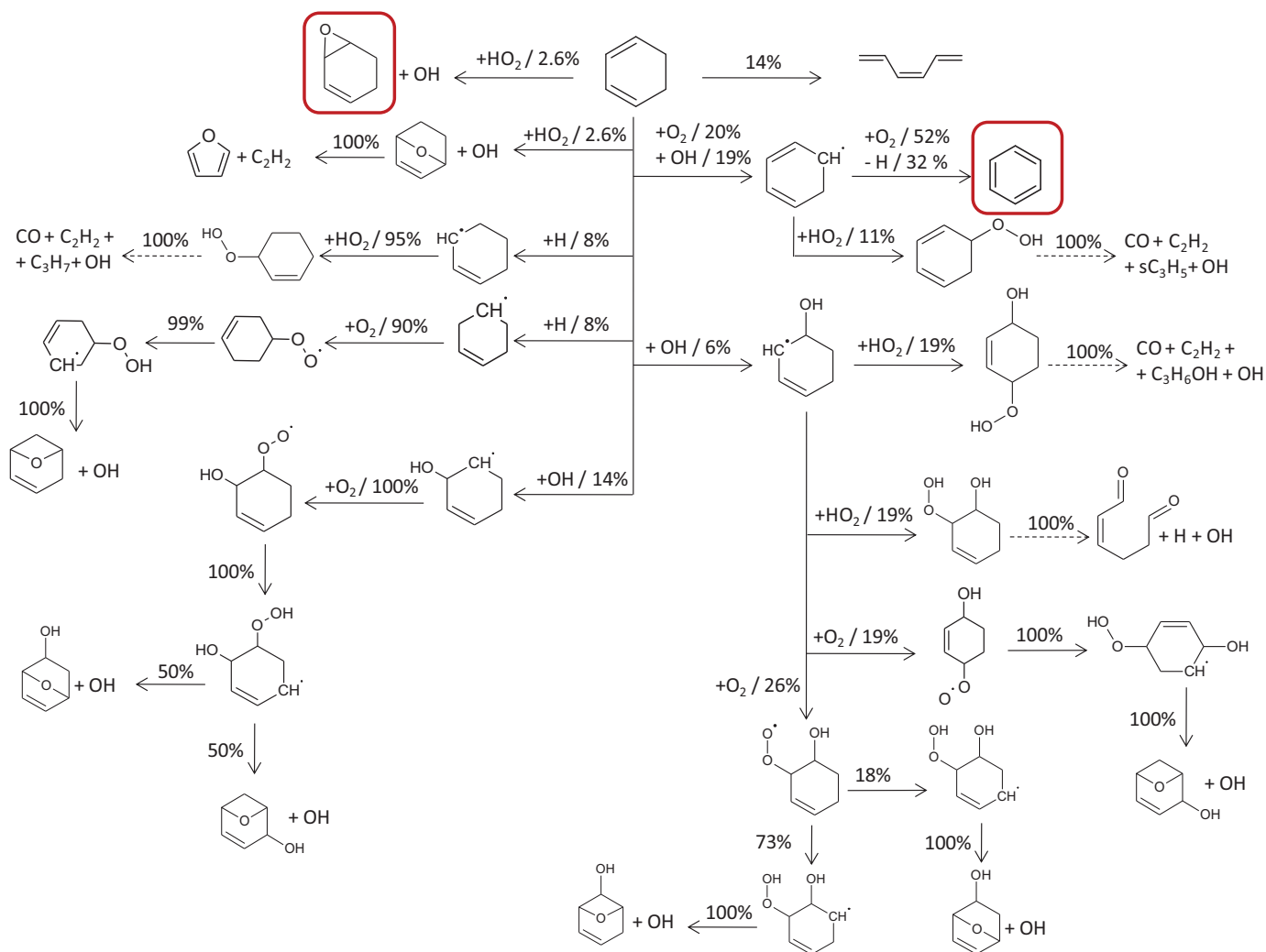


Fig. 8. Reaction flux analysis for 1,3-CHD combustion; conditions: 727 K, 10 bar, residence time of 25 ms corresponding to 25% 1,3-CHD conversion. Given reaction fluxes (numbers associated with arrows) are relative to the consumption of a given species and are expressed in percent. Species framed in red are the one experimentally measured by Lemaire et al. [19]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

simulations: H-abstractions, a pericyclic reaction and radical additions onto 1,3-CHD. H-atoms are abstracted from 1,3-CHD through bimolecular initiation with O_2 (20%) and H-abstractions by OH radicals (19%). This represents the main consumption flux under these conditions, yielding cyclohexadienyl radical. The latter radical mostly leads to the formation of benzene along with H-atoms, by β -scission (32%), and HO_2 radicals through oxidation reaction with O_2 . The entire benzene profile simulated in Fig. 4 comes from this reaction route. A minor decomposition pathway involving combination of cyclohexadienyl with HO_2 is also highlighted in the simulations. The hydroperoxide formed in this termination step reacts by initial O–O bond fission, yielding an alkoxy radical (+ OH) that ultimately decomposes into small species by successive β -scissions and isomerisations.

20% of the total consumption flux of 1,3-CHD occurs through OH addition onto the ring yielding two adducts: a resonance stabilized radical and a secondary alkyl radical. The alkyl radical is consumed by an initial addition on O_2 , followed by an isomerisation of the peroxy radical formed that ultimately leads to the formation of cyclic ethers. The other (allylic) adduct also reacts by addition on O_2 , but its resonance stabilization leads to additional decomposition pathways, involving combinations with HO_2 radicals. Hydroperoxides formed in combinations with HO_2 all follow

a similar decomposition pattern: initial O–O bond fission and successive β -scissions and isomerisations of the alkoxy radicals and its isomers. The $\text{ROO}\cdot$ radicals produced in the OH-addition initiated pathways react by following a classic low-temperature combustion chemistry scheme: $\text{ROO}\cdot \rightleftharpoons \cdot\text{QOOH} \rightarrow \text{cyclic ether} + \text{OH}$. All the cyclic ethers produced in these pathways bear a hydroxyl group.

H-atom additions account for 16% of the total consumption flux of 1,3-CHD. It is interesting to note that H-atoms are mostly produced by the C–H β -scission of cyclohexadienyl, with this reaction representing 62% of the total flux of H-atom production. The addition of H-atoms on 1,3-CHD leads to two radicals with and without resonance stabilization. Similarly to OH-addition adducts, the resonance stabilized cyclohexenyl radical mostly reacts with HO_2 , yielding a hydroperoxide, while the non-allylic cyclohexenyl is mainly involved in a peroxy radical formation through additions on O_2 . The latter pathways explain the formation of 6-oxabicyclo[3.1.1]hept-2-ene, which is represented in Fig. 6.

HO_2 additions, written in a single step yielding cyclic ether + OH, represent 5% of the total consumption flux of 1,3-CHD. The simulated profile of 1,2-epoxycyclohex-3-ene in Fig. 4 is explained by this mechanism as illustrated by the red frame in Fig. 8.

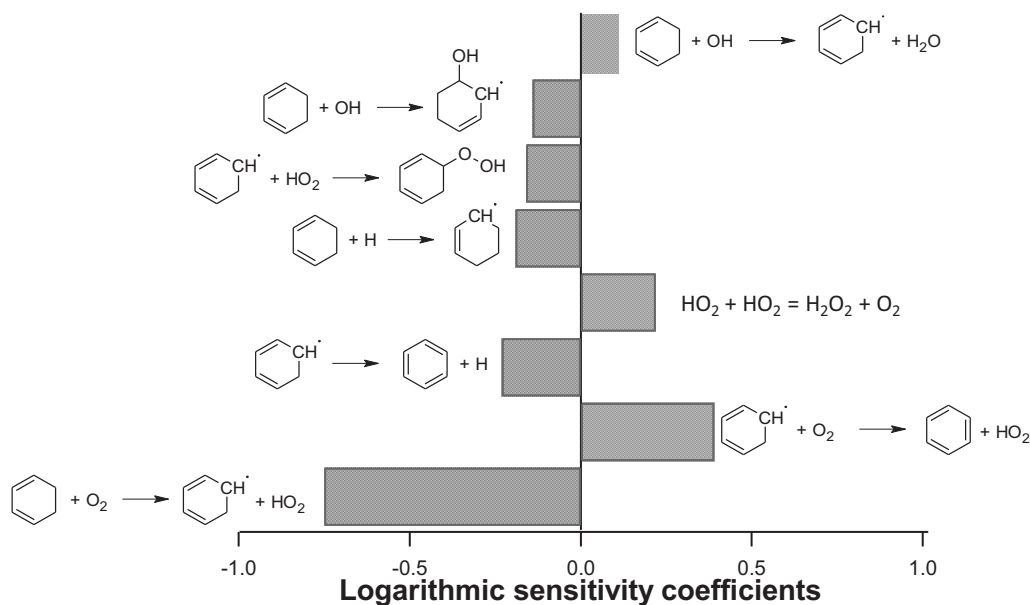


Fig. 9. Logarithmic sensitivity coefficients (see text) for RCM ignition delay time with respect to reaction rate parameters, computed for 1,3-CHD in air at $\varphi = 1$, $T = 727$ K, and $P = 10$ bar.

Under the conditions of Fig. 8, the formation of 1,3,5-hexatriene accounts for 14% of 1,3-CHD total consumption. The pericyclic reaction $1,3\text{-CHD} \rightleftharpoons 1,3,5\text{-HT}$ therefore represents a non-negligible part of the total flux of consumption of 1,3-CHD. The fate of 1,3,5-HT will be further discussed in Section 4.

Figure 9 presents ignition delays sensitivity coefficients obtained through a brute force sensitivity analysis performed for the conditions of Lemaire et al. [19]. Sensitivity coefficients were calculated using

$$\ln S_i = \frac{\ln\left(\frac{\tau_+}{\tau_-}\right)}{\ln\left(\frac{2}{0.5}\right)}$$

where τ_+ and τ_- are the computed ignition delays corresponding to a rate constant i multiplied by a factor 2 and 0.5, respectively. Simulated ignition delay times were determined with a temperature increment of 400 K.

Among all sensitive reactions depicted in Fig. 9, only the dismutation reaction between two HO_2 radicals belongs to the core reaction mechanism. The positive sensitivity coefficient for this reaction denotes an inhibiting effect on the ignition delay. This is not surprising as HO_2 is the main oxygenated radical at these temperatures and this termination consumes two of them while yielding two molecules. It can be noted that for the conditions of Fig. 8, the computed ignition delay is 48 ms. From 0 to 42 ms, the ratio of mole fractions of HO_2 to OH is $\approx 10^3$, from 42 to 47.6 ms this ratio is still about 100. All the other sensitive reactions pertain to the primary mechanism of 1,3-CHD. The most sensitive reaction is the bimolecular initiation that features a strong promoting effect on the reactivity. This reaction creates cyclohexadienyl and HO_2 radicals, and it can be seen in Fig. 9 that three reactions involving cyclohexadienyl appear in the sensitivity analysis. The oxidation reaction is found to have an inhibiting effect on ignition delay because it leads to stable benzene and HO_2 and is in competition with two other reactions that promote reactivity. These two reactions are the combination of cyclohexadienyl with HO_2 that leads to a branching agent, and the C–H β -bond scission that produces benzene and active H-atoms. H-abstraction by OH radical from 1,3-CHD has an inhibiting effect on the ignition delay. This is explained by the consumption of an active OH radical that leads to the less reactive cyclohexadienyl radical. OH and H-atom

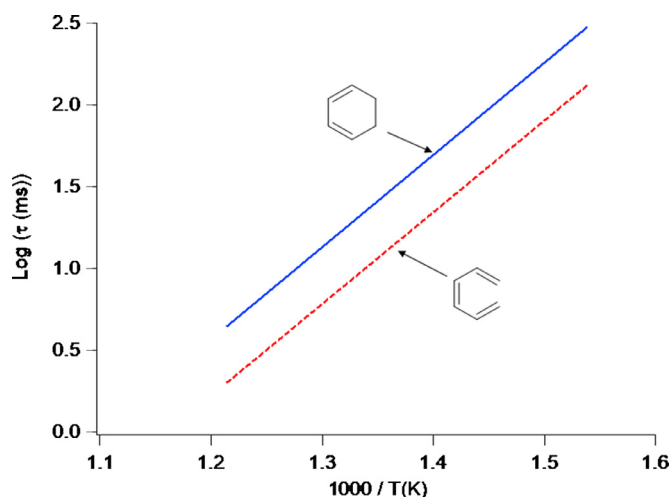


Fig. 10. Ignition delays of 1,3-CHD/ O_2 /inert (solid line) and cis-1,3,5-HT/ O_2 /inert (dashed line) mixtures in a RCM ($\Phi = 1$) at ≈ 12.8 bar.

additions on 1,3-CHD are found to promote ignition delays because they yield resonance stabilized radicals that further react by combination with HO_2 . These combinations lead to the formation of hydroperoxides that are degenerate branching agents. The sensitivity analysis presented here shows that the most sensitive reactions for the auto-ignition delays of 1,3-CHD involves several reactions that are rather uncertain because of the lack of elementary kinetic data. Further theoretical and experimental studies would be welcomed to refine the level of knowledge on this system.

4. Simulations of the auto-ignition of 1,3,5-hexatriene in a RCM

As already mentioned, no experimental data for 1,3,5-hexatriene combustion was found in the literature. Therefore, it is currently impossible to validate the detailed kinetic model for the prediction of 1,3,5-HT ignition delays. However, we performed simulations of its auto-ignition under the same experimental conditions that the one reported by Lemaire et al. [19]. Figure 10

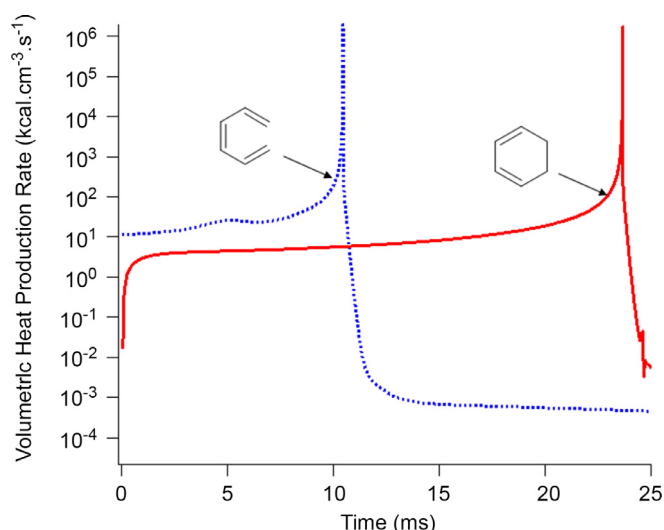


Fig. 11. Volumetric heat production rates simulated during the combustion of 1,3-CHD and 1,3,5-HT in closed homogeneous reactor. Simulations performed at a temperature of 746 K, a pressure of 12.9 bar, an equivalence ratio of 1, and 2.56% fuel / 61.6%N₂ / 15.4%Ar mixtures.

compares these simulated ignition delay times with that of 1,3-CHD.

Ignition delays of *cis*-1,3,5-HT are lower than those of 1,3-CHD by a factor of 2 from 650 to 825 K. At first glance, this result is surprising because 1,3,5-HT does not feature any favourable unimolecular or bimolecular initiation reaction compared to 1,3-CHD. The shorter ignition delays simulated for 1,3,5-HT are in fact explained by its reaction mechanism and the use of an adiabatic reactor for the simulations. The preponderant consumption flux of 1,3,5-HT involves the formation of 1,3-CHD that, in turn, is decomposed through the fluxes presented in Fig. 8. As shown in the potential energy surface in Fig. 1, the cyclisation of *cis*-1,3,5-HT into 1,3-CHD is exothermic by 11.4 kcal mol⁻¹ at 298 K. Therefore, the formation of 1,3-CHD from 1,3,5-HT leads to an increase of temperature which contributes to the shorter ignition delays observed in Fig. 10.

In order to verify this assumption, we performed a simulation with a modified detailed chemical kinetic model, where the enthalpy of formation of 1,3,5-HT is set equal to that of 1,3-CHD. In this mechanism, each reversible reaction of the sub-mechanism of 1,3-CHD and 1,3,5-HT combustion was explicitly written as irreversible in the forward and reverse directions. The MECHMOD [56] program was used to create the irreversible set of reactions. Using this modified mechanism to simulate the auto-ignition of 1,3-CHD and 1,3,5-HT in the RCM conditions presented in Fig. 10, led to similar ignition delay times for both fuels. The simulation results are given in Figure S2 of the supplemental material. Figure 11 presents the volumetric heat production rates (VHPR) taking place during the combustion of 1,3-CHD and 1,3,5-HT in an RCM.

VHPR simulations presented in Fig. 11 shows that the combustion of 1,3,5-HT right away produces 11 kcal cm⁻³ s⁻¹ while 17 ms are needed to reach the same level of VHPR in 1,3-CHD combustion. The exothermic cyclization of 1,3,5-HT into 1,3-CHD, which is the main consumption pathway of the 1,3,5-hexatriene, explains this behaviour. The peaks in VHPR profiles correspond to ignition of the mixtures.

The reaction flux analysis demonstrates that the main consumption pathway of 1,3,5-HT involves its cyclization into 1,3-CHD. Figure 12 depicts the relative rates of consumption of 1,3,5-HT as a function of its conversion.

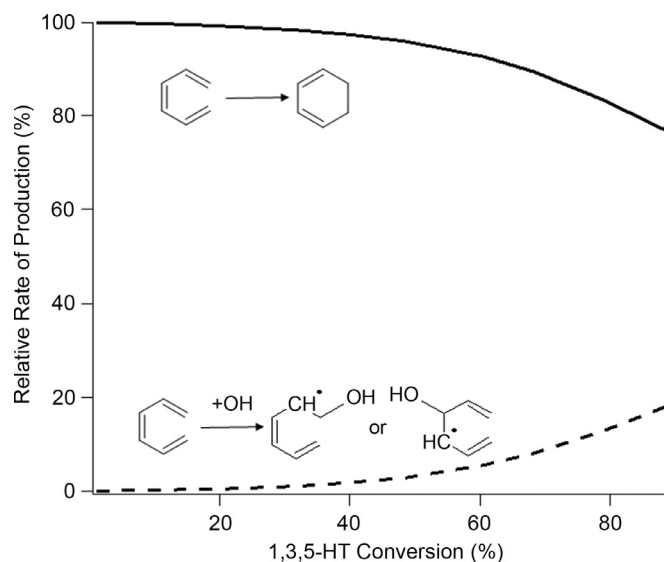


Fig. 12. Relative rates of production (%) of the main decomposition pathways of 1,3,5-HT as a function of its conversion (%). Simulation of 2.56% 1,3,5-HT in air performed at 727 K, at a pressure of 10 bar and an equivalence ratio of 1.

It can be observed that the cyclization of 1,3,5-HT into 1,3-CHD is preponderant up to 90% of conversion. Other consumption pathways remain negligible for conversions below ~60%. Once 60% of conversion of 1,3,5-HT is reached OH-addition pathways start to play a role on the decomposition pathways of the fuel. The two resonance stabilized adducts are formed in similar proportions in the OH addition processes and mainly react by addition on O₂ or internal H-atom transfer and follow the classic low temperature reaction scheme. The oxidation mechanism of 1,3,5-HT therefore involves an induction period leading to the formation of 1,3-CHD. The decomposition of 1,3-CHD will increase the number of radicals that will be further able to react with 1,3,5-HT, mostly by additions.

Our simulations predict that the reactivity of pure 1,3-CHD is lower than that of pure 1,3,5-HT under adiabatic conditions. The kinetic analyses of 1,3,5-HT performed with the reaction mechanism provide a credible explanation for this result. However, experimental data on the combustion of 1,3,5-HT will be necessary to confirm this prediction.

5. Summary and conclusions

A detailed chemical kinetic model for the low-temperature combustion of 1,3-cyclohexadiene was developed in this work, for the first time in the literature. A large number of thermodynamic data was calculated using theoretical chemistry tools. It was demonstrated that *cis*-1,3,5-hexatriene is a primary product of 1,3-CHD mechanism and its comprehensive mechanism was included in the secondary mechanism of 1,3-CHD. Comparisons between simulations and two sets of experimental data measured in different RCMs for 1,3-CHD showed a good agreement between ignition delays and products speciation. RCM experiments by Tanaka et al. demonstrated that 1,3-CHD ignition delays measured at T_c = 827 K, P_c = 4.16 MPa, and for an equivalence ratio of 0.4, were shorter than those of cyclohexane and cyclohexene. The kinetic analyses performed with our validated model show that this relatively surprising high reactivity could be explained by favourable bimolecular initiations with O₂, yielding HO₂ and cyclohexadienyl radical, which in turn leads to the formation of benzene and H-atom by β-scission or to the production of benzene and HO₂ by oxidation with O₂. The kinetic parameters of these reactions were found

to be the most sensitive on the simulated auto-ignition delays of 1,3-CHD in a RCM.

Exploratory simulations of *cis*-1,3,5-hexatriene low-temperature combustion in a RCM showed that this fuel surprisingly features a higher reactivity than 1,3-CHD, although it has only vinylic H-atoms. Using our detailed chemical kinetic model, it was demonstrated that *cis*-1,3,5-HT predominantly decomposes by cyclization into 1,3-CHD. Since this reaction is exothermic, simulations in an adiabatic reactor leads to shorter ignition times because of the heat release. 1,3-CHD and 1,3,5-HT reaction mechanisms are therefore strongly interconnected.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2019.04.020](https://doi.org/10.1016/j.combustflame.2019.04.020).

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